

Elastic Impression materials

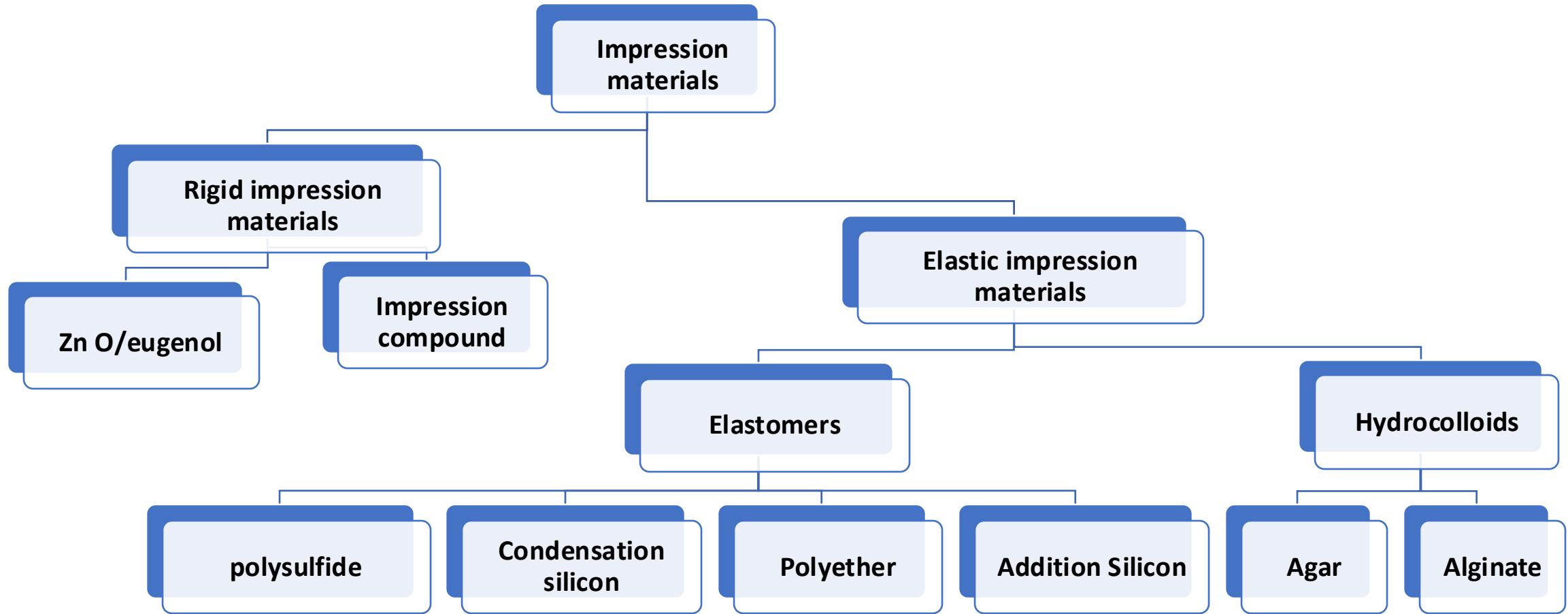
By

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Classification of impression materials



Classification of Dental Impression Materials

Setting Mechanism	Rigid (inelastic)	elastic
Chemical reaction (irreversible)	Zinc oxide– eugenol	-Alginate -Polysulfide -Polyether -Condensation silicone -Addition silicone
Thermally induced physical reaction (reversible)	Impression compound	Agar

What is the difference between elastic and rigid impression materials?

- The impression materials after setting can be rigid (inelastic) or elastic.
- A set rigid material is highly resistant to flexure, and it fractures suddenly under stress (like upon removal from mouth), like chalk as : ZOE impression paste, impression plaster, and impression compound.
- *elastic impression materials* means that the material is flexible and can be deformed and still return to its original form to a great extent when unstressed i.e. after removal from mouth. Examples include: agar, alginate, and elastomers.

- **Q: Can we use rigid impression materials to take impressions for intraoral areas with undercuts ?**
- **A: No, because on setting they would be locked in undercut areas (e.g. interproximal areas) and upon removal they would be permanently deformed and fractured**

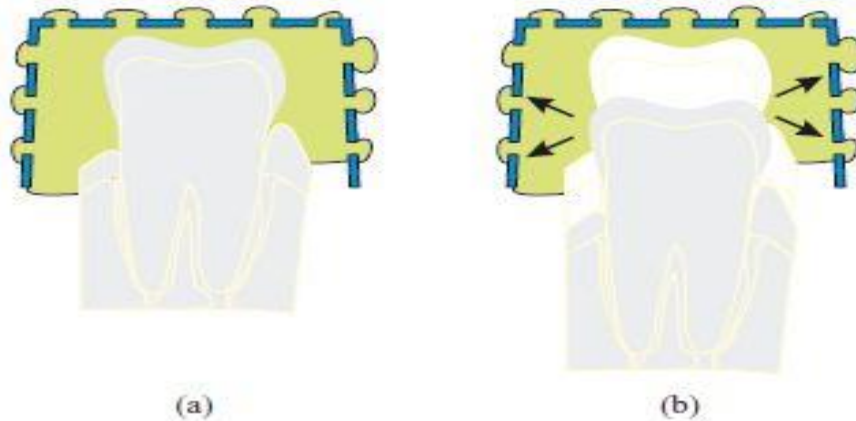


Diagram illustrating how an impression material is placed under stress during removal from an undercut area. (a) Impression in place before removal. (b) During removal

I-Elastic impression materials

A-Hydrocolloids

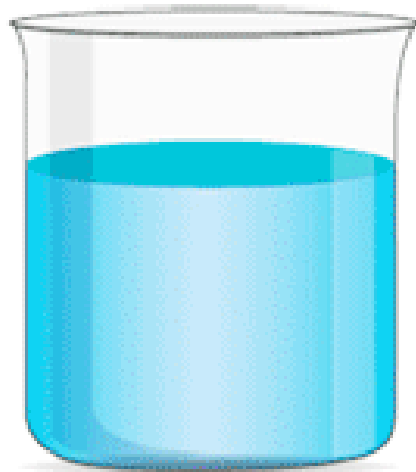
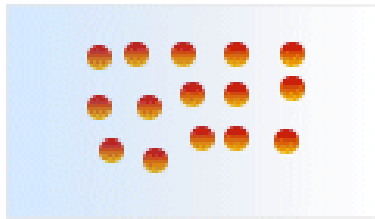
What are colloids?

- Substances are called colloids when **they consist of two or more phases** with the units of at least one of the phases having a dimension slightly greater than simple molecular size
- The range of size is usually recognized as being **approximately 1 to 500 nm** measured in maximum dimension. However, there are many variations between different references about determining the range of particle size of the colloidal phases
- Colloid is a **Heterogenous mixture** with dispersed phase having **particle size intermediate between true solution and suspension**

Types of solutions

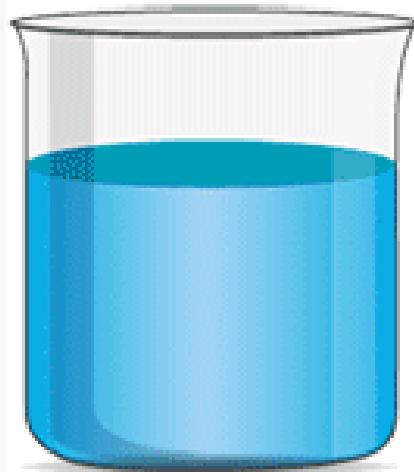
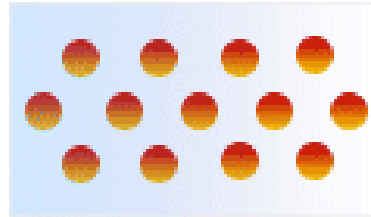
True solution

Particles size
less than 10^{-7} cm



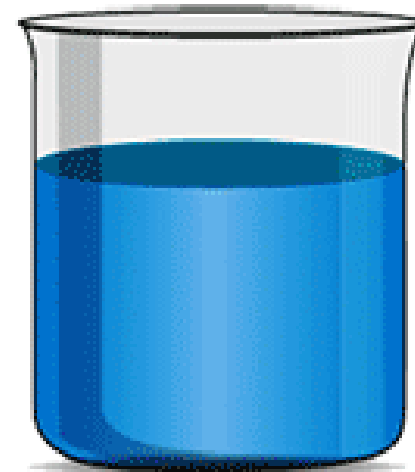
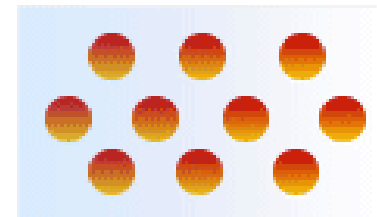
Colloidal solution

Particles size
 10^{-7} to 10^{-5} cm



Suspension

Particles size
greater than 10^{-5} cm



Important notes about colloids:

1-True solution = Single phase

2-Colloid = two separate phases = Dispersed phase + Dispersion phase

3-When dispersion phase of a colloid is water it is called Hydrocolloid

4-Colloids can exist in two states sol or gel

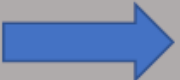
5-Colloids can experience sol to gel or vice versa transformation under certain conditions and this transformation could be reversible or irreversible.

6- In sol state the colloid exists as a viscous liquid

7- Gel exists as an elastic solid or semisolid or jellylike.

Quiz :

If a hydrocolloid is an impression material so it must be inserted in the mouth in ----- state to be sufficiently fluid to accurately catch intraoral details, and left in the mouth sufficient time to be converted into ----- state to develop enough elastic properties to be removed from mouth without distortion

- Gel of hydrocolloids = hydrogel , sols of hydrocolloids = hydrosol
- Hydrogels are hydrophilic (water loving) so if submerged in water they will absorb or imbibe water , *imbibition*  *swelling* and change in dimensions
- On the other hand, Hydrogels could lose water if left in dry air by **evaporation** in a process called *syneresis* , which results in *shrinkage* and change in physical dimensions.

Alginate (irreversible hydrocolloids)

- **Alginate** : one of the most widely used impression materials.
- The wide use due to advantages :
- (1) the ease of mixing and manipulation (2) the minimum equipment; (3) the flexibility of the set impression; (4) accuracy (5) low cost.

Disadvantages : 1- low tear strength

2-low elastic recovery

3- poor dimensional stability due to syneresis and

imbibition

Applications:

- 1- primary impressions
- 2- making of study models
- 3- making athletic mouth protectors

Practical



Presentation:

- Powder packaged in **bulk or in pre-weighed individual packets**
- **The bulk material** is packaged in a sealed plastic container; in a sealed metal can, or in foil packets

- The pre-weighed packages are constructed of plastic and metal foil and contain enough material for a single full-arch impression.
- These packages minimize moisture contact and extend the storage life.
- A plastic scoop for powder
- A plastic cylinder is supplied for measurement of the water
- Mixing by :
 - Manually by A wide-bladed, stiff spatula to mix the powder and water.
 - Or automatically by Mechanically driven-mixers.

Practical



Practical



Figure 2. Manual alginate impression material mixing, water and powder. Courtesy Prof M.C.



Figure 3. Manual alginate impression material mixing. Alginate on dental impression tray. Courtesy Prof M.C.



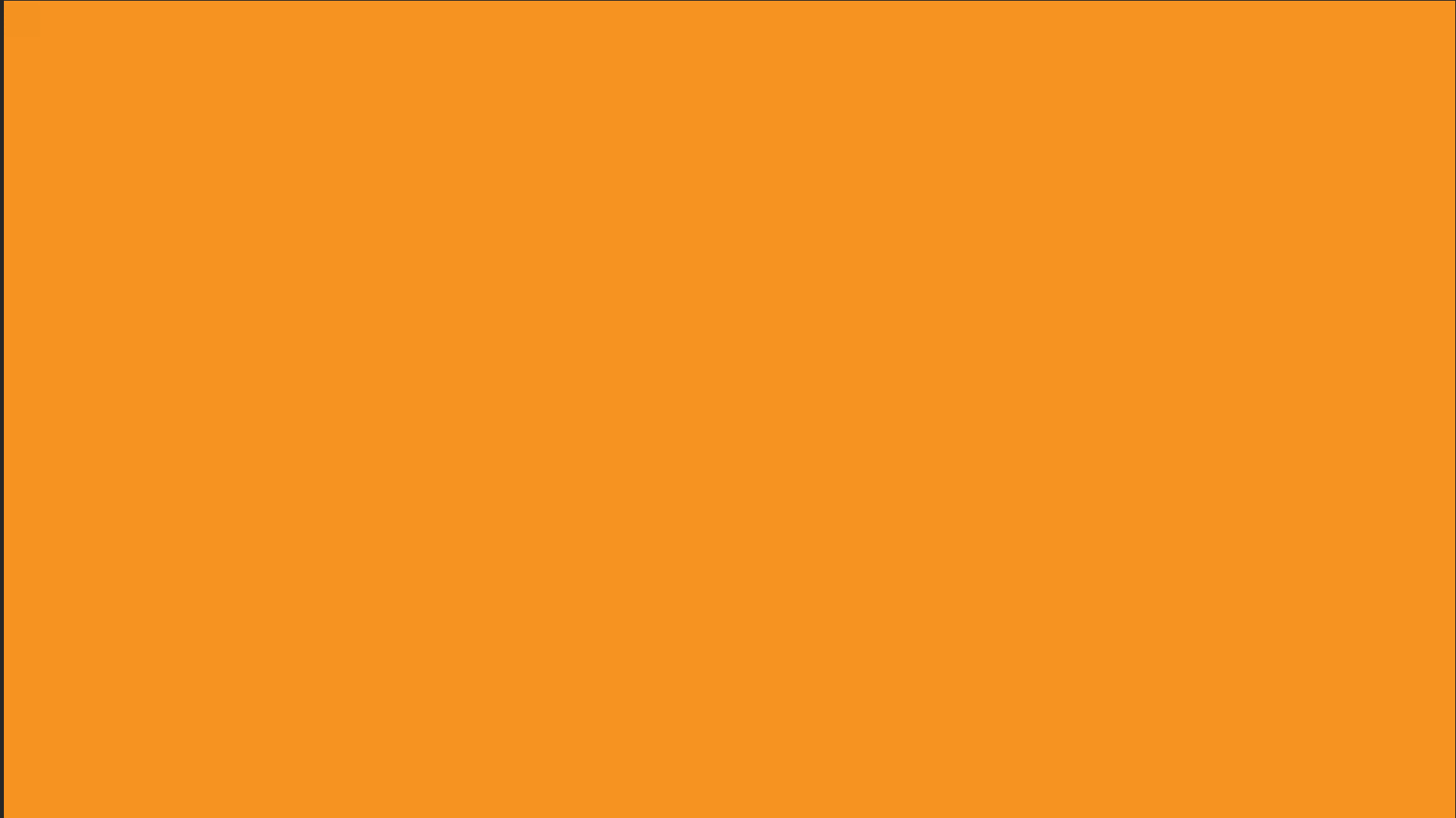
Figure 1. Alginate automixer Cavex®, for alginate mixing.

Source of figures paper :

Review

Alginate Materials and Dental Impression Technique: A Current State of the Art and Application to Dental Practice

Gabriele Cervino ¹, Luca Fiorillo ¹, Alan Scott Herford ², Luigi Laino ³, Giuseppe Troiano ⁴,
Giulia Amoroso ¹, Salvatore Crimi ⁵, Marco Matarese ¹, Cesare D'Amico ¹,
Enrico Nastro Siniscalchi ¹ and Marco Ciccù ^{1,*}



Practical

———Since 2009———

Composition :

- **Sodium or potassium alginate salt** : To dissolve in water
- **Calcium sulfate**: To react with dissolved alginate to form insoluble calcium alginate
- **Sodium phosphate** : To react preferentially with calcium sulfate and act as a retarder
- **Diatomaceous earth or silicate powder**: To control consistency of mix and flexibility of impression
- **Potassium sulfate or potassium zinc fluoride**: To counteract the inhibiting effect of alginate on the setting of gypsum material
- **Organic glycol** : To coat the powder particles to minimize dust, to produce dust free alginate, due to health hazards of dust inhalation
- **Pigments**: To provide color
- **Quaternary ammonium compounds or chlorhexidine** : self-disinfectant

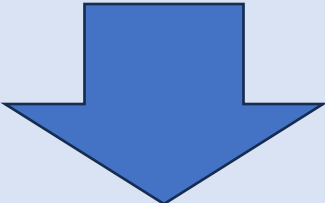
When alginate powder is fluffed  the diatomaceous earth, with fine porous silica particles will become **airborne** as the cover is removed.

Long-term exposure by inhalation can cause Silicosis and pulmonary hypersensitivity.

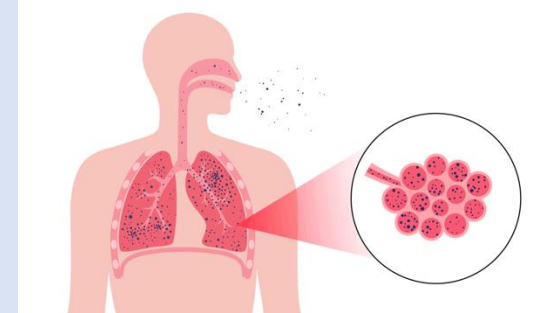
**So,- you have to wear mask before removing cover
-Put water in bowl before powder**

Manufacturers have introduced a **"dustless" alginate**


polyethylene glycol or polypropylene glycol in the alginate powder


agglomerate the particles

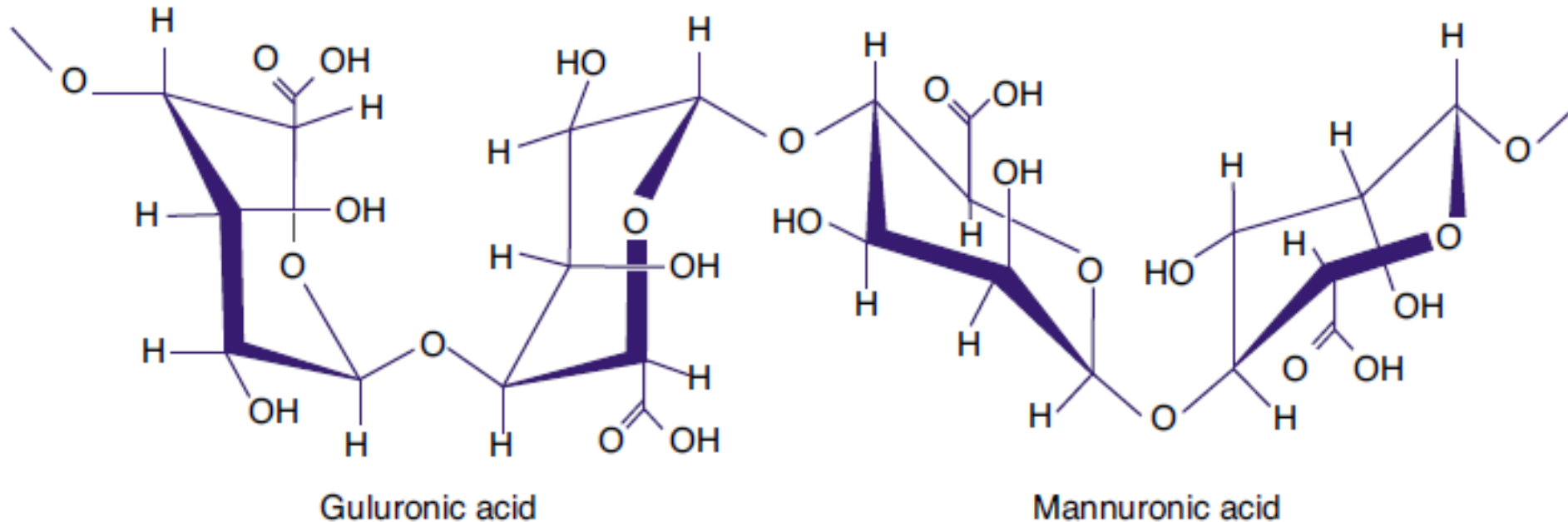
powder becomes **denser and less able to become airborne.**



The setting of alginate hydrocolloid is a process of alginic acids with calcium ions.

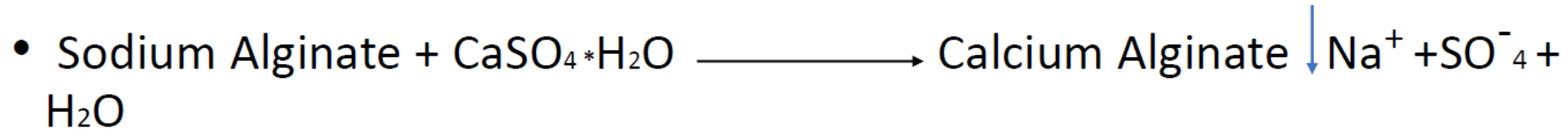
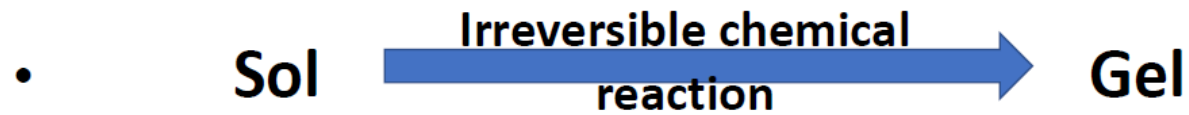
crosslinking

The alginic acid, which is extracted from brown seaweed, is a linear copolymer of ***β -D-mannuronic acid and α -L-guluronic acid***



Setting mechanism and manipulation:

- water is mixed with the alginate powder to form a smooth viscous mass which becomes an **irreversible gel** a few minutes after mixing as follows:



- As long as any sodium phosphate is present, it prefers to react with the soluble calcium ions.



After all the sodium phosphate has been reacted and consumed



sodium alginate reacts with the remaining calcium ions and **calcium alginate** precipitates.



- sodium phosphate therefore is called a **retarder**.



- **The manufacturer controls the time of setting by the amount of sodium phosphate present in the alginate powder.**

- The calcium alginate precipitates into a fibrous network; water occupies the in between capillary spaces. **This type of structure is called a gel.**

- According to retarder conc.
There are two types of alginate:

Fast setting (1.5-3 min) and
regular setting (3-4.5 min)

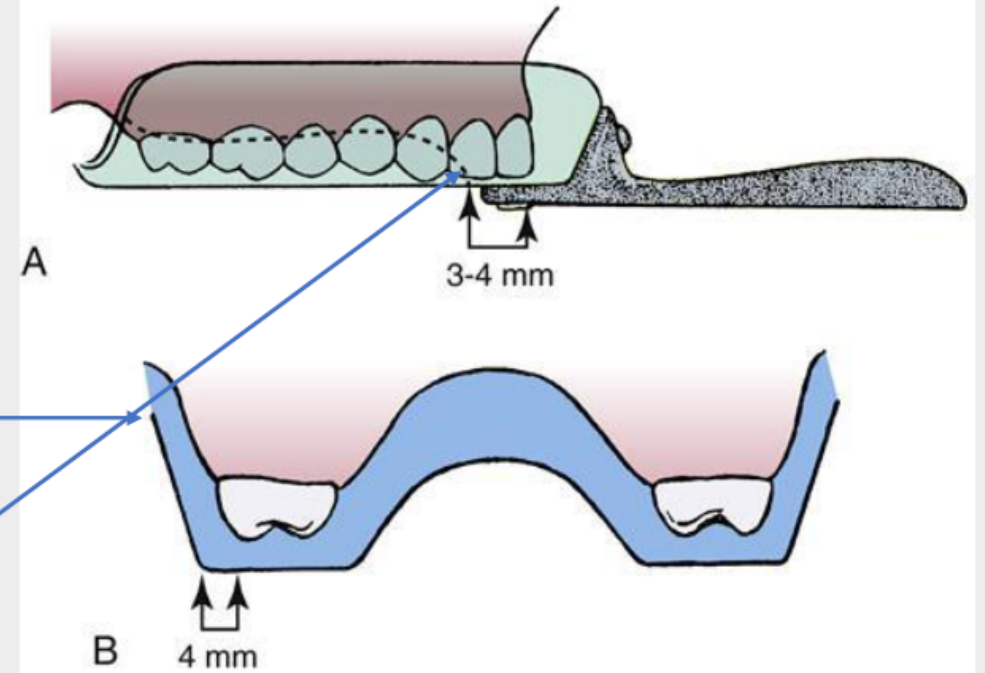


Practical

• Steps of manipulation:

• 1- Tray selection:

- **Must completely cover all teeth + maxillary tuberosity or mandibular retromolar pad**
- Be 4 mm wider than the most apical portion of the alveolar process at the upper molar region
- Cover the anterior teeth with the incisors contacting the flat arch portion of the tray about 4 mm from the raised palatal part of the upper tray
- Be 4 mm wider than the buccal and lingual positions of the lower posterior and the labial and lingual positions of the lower anterior teeth



Correct tray selection for a maxillary alginate impression. A, Side view. B, Posterior view. (Modified from Meldrum RJ, Johnson RA, Cheney EA: *Preclinical Orthodontics* 621/623, Ann Arbor, 1980, School of Dentistry, University of Michigan.)

<https://pocketdentistry.com/8-impression-materials/>

- **2-Dispensing the Alginate:**
- The recommended amount of water is **measured into rubber bowl** using **the manufacturer's liquid measuring set**.
- The **water temperature** should be at room temperature (22° to 23° C).
- a full measure free from large voids by **The manufacturer's powder scoop** is taken .
- Finally, **the excess powder** is scraped off with a spatula.
- The powder is put into **the rubber bowl** that contains the water.



- **3-Mixing the Alginate:**

- 1- Using a **stiff, wide-bladed spatula**, the alginate powder is stirred with water.

- 2- The mix is spatulated **vigorously** against the side of the rubber bowl

- 3- until a **smooth, creamy mix** that is **free of voids** results.

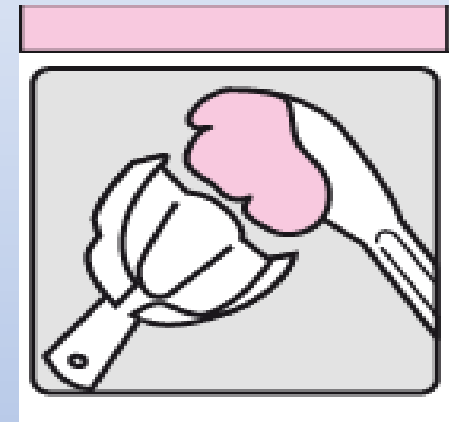
- 4- Cold water extend mixing time

Insufficient spatulation results in a grainy mix.



Practical

- 4-Loading the Tray:
- The spatula is used to transfer the alginate from the bowl to the tray.
- The tray is filled to the top , make sure that there are no voids.
- A useful method is to start at the posterior to anterior by adding more and more to the posterior.



- 5- Taking impression:
- A- insert loaded tray in rotational movement
- B- hold tray in position till 2-3 minutes after alginate loses its tackiness

NOTE: you take small piece of excess alginate and put it on your gloves and wait about 2-3 minutes after it loses its tackiness to remove tray

- C- remove in a **snap sharp rapid motion** to increase alginate tear strength
- D- alginate imp. Is taken using perforated stock tray **why? For mechanical interlocking**
- E- during taking imp. **Do not move tray or exert much pressure why?** Because alginate in contact with tissues sets first due to higher temp., so any pressure on gel would result in stresses which would be released later → distortion of imp.

properties:

Practical

- 1-Mixing and Setting Times:
- Because the setting occurs as a result of an irreversible chemical reaction, an **increase in the temperature of the water shortens the working and setting times.**
- proportions of powder and water also affect the setting times. **Thinner mixes increase setting time (take care: thin mix causes decrease in tear strength).**
- **If you want to delay setting , reduce temp. of water but do not increase water ratio (do not use thin mix)**

- 2-Mechanical properties :
- - Elastic recovery:
- The ANSI-ADA specification: at least 95% elastic recovery

Alginate is The most flexible elastic impression

alginate imp. is flexible but not perfectly elastic (i.e. viscoelastic)

So, Remove by sharp rapid snap from patient mouth to:
decrease permanent deformation

Tear strength



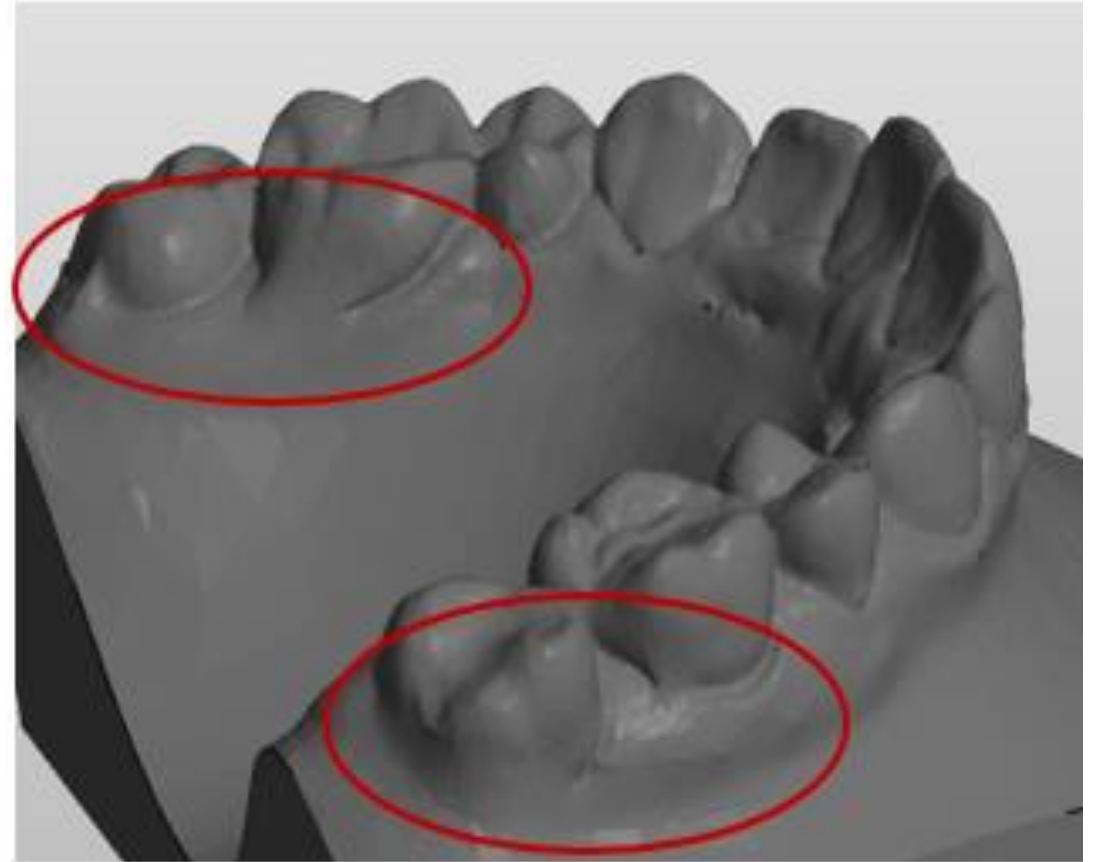
- Definition : The resistance of impression material to fracture (tear) when subjected to a tensile force
- •Hydrocolloids have relatively low tear strength, so:-
 - Thickness must be from 4-6 mm “look at tray selection”
 - During removal from patient mouth, remove by sharp rapid snap

Teared impression



<https://support.clearcorrect.com/hc/en-us/articles/216946298-Impressions-Problems#9>

Inaccurate cast



3- Dimensional stability:



Syneresis and imbibition problems

- Storage in either **air or liquid** results in high dimensional changes due to syneresis (in air) and imbibition (in liquid).
- You have to pour alginate imp. as soon as possible after disinfection (about 10 min) **(option number 1),**
- if that is not possible store in 100% relative humidity for shortest time or wrap it by wet towel **(option number 2).**
- So, the most appropriate procedure is to pour alginate after disinfection immediately (10-15 min.)

- 4-Dimensional accuracy:
 - The impression material must record the detail of the oral tissues, but this detail also **must be transferred to the model or die.**
 - the minimum ability must be to transfer a line only 0.075 mm wide to a gypsum model or die material.
- Alginate undergoes slight shrinkage during setting **due to agglomeration of dispersed phase (like Agar)**



Practical

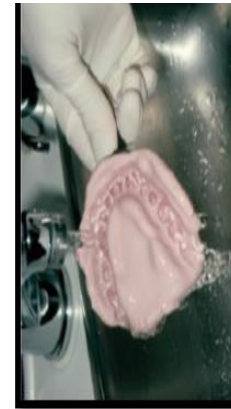


• 4- COMPATIBILITY WITH GYPSUM

- After imp. rinse well in cold water to remove saliva and any blood, then disinfect.
- Next after disinf. , all free surface water should be removed before pouring a gypsum model.
- Take care : 1- Saliva , blood interfere with gypsum setting,
- 2- free accumulated water dilute gypsum and produces soft chalky gypsum surface.
- 3- Don t leave the cast too long in contact with impression after setting (set gypsum will absorb water from impression that results in cast with chalky surface).
-

D. Disinfection

- Wear all your PPE (gloves ,long sleeve gown , head cover and face-shield) during imp. Disinfection
- Rinse impression well with cold running water to remove blood and saliva
- Choose disinfectant with short exposure time (about 10 min)
- Sodium hypochlorite and iodophors could be used
- Follow manufacturer' instructions (time, concentration and type)
- Avoid use of Glutaraldehyde
- Rinse well (for minimum 30 sec) before gypsum pouring



References

- **References and photos are from:**
- **1- Craig's 14th edition properties of restorative dental materials**
- **2- pocket dentistry website**
- **3- Philips science of dental materials**
- **4- text book of applied dental materials , faculty of dentistry , Cario university**



• **THANK YOU**